CEGE: Conversational Emotion Graph Evolution*

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Abstract

Emotion Recognition in Conversation (ERC) is an important task for enabling empathetic and socially-aware dialogue systems. Existing methods often rely on sequential encoders or static graph structures that fail to capture the evolving nature of emotional influence across dialogue turns. We propose a dynamic temporal graph neural network that models conversations as evolving graphs, continuously updating utterance representations and influence edges as the dialogue unfolds. Our framework primarily focuses on text-based ERC, with potential extensions to audio and video modalities if time permits.

1 Introduction

Emotion Recognition in Conversation (ERC) is crucial for building empathetic dialogue systems. Current models often treat conversations as linear sequences, but real dialogues exhibit non-linear emotional dependencies:

- Emotional states can span multiple turns and speakers.
- Some utterances have a disproportionate influence on later emotions.
- Influence strength changes dynamically as the dialogue progresses.

Most existing graph-based ERC models use static graphs, failing to update relationships as the conversation evolves. We propose **Conversational Emotion Graph Evolution (CEGE)**, a dynamic graph-based ERC model that continuously updates utterance representations and their influence edges over time. CEGE aims to better capture the temporal evolution of emotional influence across participants.

*https://github.com/farzanashaju/cege

2 Related Works

Emotion Recognition in Conversation (ERC) has advanced from sequential models to graph-based and, more recently, dynamic frameworks. DialogueRNN (Majumder et al., 2019) was an early influential model, using recurrent networks with attention to capture speaker-specific emotional dependencies. DialogueGCN (Ghosal et al., 2019) extended this idea with graph convolutional networks to model both intra- and inter-speaker relations.

While effective, such static graph structures cannot reflect the evolving influence of utterances over time. To address this, dynamic methods were introduced. DGODE (Shou et al., 2024) employed graph ordinary differential equations to capture continuous emotional dynamics, while ERC-DP (Wang et al., 2024) modeled dynamic speaker personalities. More recently, GSDNet (Shou et al., 2025) applied spectral graph diffusion to enhance temporal context tracking.

These works mark a clear shift from static to adaptive graph modeling, emphasizing the importance of capturing evolving emotional dependencies in dialogue.

3 Objective

Existing ERC approaches face several challenges:

- **Static context modeling:** Relationships between utterances remain fixed once encoded.
- **Limited influence tracking:** The changing significance of earlier utterances is not captured.

We aim to develop a dynamic graph evolution framework for ERC that models changing emotional influence between utterances and updates the conversational context over time.

4 Dataset

We evaluate our approach on the IEMOCAP dataset. IEMOCAP (Busso et al., 2008) dataset contains videos of two-way conversations of ten unique speakers, where only the first eight speakers from session one to four belong to the trainset. Each video contains a single dyadic dialogue, segmented into utterances. The utterances are annotated with one of six emotion labels, which are happy, sad, neutral, angry, excited, and frustrated.

5 Proposed Methodology

We present **CEGE: Conversational Emotion Graph Evolution**, a novel architecture that models conversations as dynamically evolving temporal graphs for Emotion Recognition in Conversation (ERC). Our approach is built from three tightly integrated components: utterance encoding, dynamic temporal graph construction, and emotionally-aware classification. Each is designed to capture the evolving nature of emotions and speaker relationships as dialogue unfolds.

5.1 Problem Definition

Let there be M speakers/parties $p_1, p_2, \ldots, p_M$ in a conversation. The task is to predict the emotion labels (*happy, sad, neutral, angry, excited, frustrated, disgust, and fear*) of the constituent utterances $u_1, u_2, \ldots, u_N$, where utterance u_i is uttered by speaker $p_{s(u_i)}$, with s being the mapping between utterance and index of its corresponding speaker. We also represent $u_i \in R^{D_m}$ as the feature representation of the utterance, obtained using the feature extraction process described below.

5.2 Utterance Encoding

We employ convolutional neural networks (CNN) for textual feature extraction. We obtain n-gram features from each utterance using three distincxt convolution filters of sizes 3, 4, and 5 respectively, each having 50 feature-maps. Outputs are then subjected to max-pooling followed by rectified linear unit (ReLU) activation. These activations are concatenated and fed to a 100 dimensional dense layer, which is regarded as the textual utterance representation. This network is trained at utterance level with the emotion labels.

5.3 Sequential Context Encoding:

Since, conversations are sequential by nature, contextual information flows along that sequence. We feed the conversation to a bidirectional gated recurrent unit (GRU) to capture this contextual information: $g_i = \overleftrightarrow{\mathrm{GRU}}_S(g_{i(\pm 1)}, u_i)$, for $i = 1, 2, \ldots, N$, where u_i and g_i are context-independent and sequential context-aware utterance representations, respectively.

Since the utterances are encoded irrespective of its speaker, this initial encoding scheme is speaker-agnostic.

5.4 Dynamic Temporal Graph Construction

A conversation having N utterances is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \mathcal{W})$, with vertices/nodes $v_i \in \mathcal{V}$, labeled edges (relations) $r_{ij} \in \mathcal{E}$ where $r \in \mathcal{R}$ is the relation type of the edge between v_i and v_j and α_{ij} is the weight of the labeled edge r_{ij}, with $0 \leq \alpha_{ij} \leq 1$, where $\alpha_{ij} \in \mathcal{W}$ and $i, j \in [1, 2, \ldots, N]$.

Temporal Memory Modules: Each node maintains a memory state $\mathbf{m}_i^t$:

$$\mathbf{m}_i^t = \mathrm{LSTM}(\mathbf{m}_i^{t-1}, \mathbf{g}_i, \mathrm{context}^t)$$

where context includes speaker states $\mathbf{Q}_s^t$ and global conversation state $\mathbf{C}^t$.

5.5 Graph Representation and Temporal Memory

A conversation with N utterances is represented as a directed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \mathcal{W})$$

where:

- $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ is the set of nodes, each node v_i corresponding to the i-th utterance.
- $\mathcal{E}$ is the set of directed edges, where each edge r_{ij} connects node v_j (source) to node v_i (target).
- $\mathcal{R}$ is the set of relation types, categorizing edges by speaker dependencies, temporal order, or learned interaction patterns.
- $\mathcal{W} = \{\alpha_{ij}\}$ is the set of edge weights, with $\alpha_{ij} \in [0, 1]$ quantifying the influence of utterance j on i.

Temporal Memory Modules Each node v_i maintains a temporal memory state $\mathbf{m}_i^t$ at time t, which summarizes the evolving emotional and contextual information for utterance i:

$$\mathbf{m}_i^t = \mathrm{LSTM}\left(\mathbf{m}_i^{t-1}, \mathbf{g}_i, \mathrm{context}^t\right)$$

where:

- $\mathbf{m}_i^{t-1}$ is the previous memory state for node i.
- $\mathbf{g}_i$ is the context-aware embedding of utterance i.
- context^t includes:
  - $\mathbf{Q}_s^t$: the speaker state for speaker s at time t, capturing emotional and interactional behavior. For utterance i, the relevant speaker is $s(i)$.
  - $\mathbf{C}^t$: the global conversation state at time t, encoding overall topic flow and emotional climate.

Graph Evolution At every new utterance (time t), the graph is updated as follows:

1. **Temporal state change** Δ_t is computed, summarizing changes in emotion, topic, and speaker patterns:
$$\Delta_t = f_{\text{analyze}}\left(\mathbf{M}^{t-1}, \mathbf{g}_t, \mathbf{Q}_s^{t-1}\right)$$
where $\mathbf{M}^{t-1}$ is the set of all node memories at $t-1$.

2. **Edge weights** are updated for each edge (i, j):
$$\alpha_{ij}^t = f_{\text{temporal}}\left(\alpha_{ij}^{t-1}, \Delta_t, \mathbf{Q}_{s(i)}^{t-1}, \mathbf{Q}_{s(j)}^{t-1}\right)$$
where f_{temporal} is a learned function (e.g., neural network or attention mechanism) that updates edge strength based on previous weight, temporal change, and speaker states.

3. **Edge pruning and addition:**
   - Edges with $\alpha_{ij}^t < \tau_{\text{remove}}$ are pruned to remove weak influences.
   - New edges are added if candidate scores (computed for plausible node pairs) exceed a threshold τ_{create}, allowing the graph to capture emerging long-range or non-local relationships.

4. **Relation type adaptation:** The set of relation types $\mathcal{R}^t$ is dynamically updated based on learned interaction patterns, enabling the model to discover new types of conversational dependencies (e.g., emotional influence, topic shift, agreement/disagreement).

5.6 Temporal Graph Convolution and Attention

The node features are passed through two temporal GCN layers at each time t:

Layer 1:

$$\mathbf{h}_i^{(1),t} = \sigma\left(\sum_{r\in R^t}\sum_{j\in\mathcal{N}_i^r}\frac{\alpha_{ij}^t}{c_{i,r}}W_r^{(1)}[\mathbf{g}_j||\mathbf{m}_j^t] + W_0^{(1)}[\mathbf{g}_i||\mathbf{m}_i^t]\right)$$

where:

- $\mathbf{h}_i^{(1),t}$ is the first-layer hidden feature for node i at time t.
- σ is the activation function, typically ReLU.
- R^t is the set of relation types at time t.
- $\mathcal{N}_i^r$ is the set of neighbors of node i with relation r.
- α_{ij}^t is the weight of the edge from node j to node i at time t, quantifying influence strength.
- $c_{i,r}$ is a normalization constant (e.g., number of neighbors of type r).
- $W_r^{(1)}$ is a learnable weight matrix for relation r in layer 1.
- $[\mathbf{g}_j||\mathbf{m}_j^t]$ denotes the concatenation of the context-aware embedding and temporal memory of neighbor j.
- $W_0^{(1)}$ is a learnable self-connection weight matrix for node i.

Layer 2:

$$\mathbf{h}_i^{(2),t} = \sigma\left(\sum_{j\in\mathcal{N}_i} W^{(2)}\mathbf{h}_j^{(1),t} + W_0^{(2)}\mathbf{h}_i^{(1),t}\right)$$

where:

- $\mathbf{h}_i^{(2),t}$ is the second-layer hidden feature for node i at time t.
- $\mathcal{N}_i$ is the set of all neighbors of node i.
- $W^{(2)}$ is a learnable weight matrix for neighbor aggregation in layer 2.
- $W_0^{(2)}$ is a learnable self-connection weight matrix for node i in layer 2.

Temporal Attention: To focus on relevant history, attention weights are computed:

$$\beta_i^t = \text{softmax}\left((\mathbf{h}_i^{(2),t})^T W_\beta [\mathbf{m}_1^t, ..., \mathbf{m}_{i-1}^t]\right)$$

where:

- β_i^t is the attention weight vector for node i at time t, indicating the relevance of each previous memory.
- W_β is a learnable weight matrix for attention scoring.
- $[\mathbf{m}_1^t, ..., \mathbf{m}_{i-1}^t]$ are the memory states of all previous utterances up to $i - 1$.
- softmax normalizes the scores into probabilities.

Temporal decay is applied to further emphasize recent context:

$$\beta_{ik}^t = \beta_{ik}^t \cdot \exp(-\lambda_{\text{decay}} \cdot (t - k))$$

where:

- β_{ik}^t is the decayed attention weight for how much utterance k should influence node i at time t.
- λ_{decay} is a hyperparameter controlling the rate of decay.
- $(t - k)$ is the temporal distance between the current and past utterance.

The context vector is then computed as:

$$\mathbf{c}_i^t = \sum_{k=1}^{i-1} \beta_{ik}^t \cdot \mathbf{m}_k^t$$

where $\mathbf{c}_i^t$ is a summary of the most relevant historical context for node i at time t, aggregating past memory states weighted by their decayed attention scores.

5.7 Training and Inference

The model is trained by minimizing the following total loss:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{reg}} \|\theta\|_2^2 + \lambda_{\text{temp}} \sum_i \|\mathbf{m}_i^t - \text{stop\_grad}(\mathbf{m}_i^{t-1})\|_2^2$$

where:

- $\mathcal{L}_{\text{CE}}$ is the **cross-entropy loss** for emotion classification. It measures how well the predicted emotion probabilities match the true emotion labels for each utterance, encouraging the model to make accurate predictions.
- $\lambda_{\text{reg}} \|\theta\|_2^2$ is the **L2 regularization term**, where θ denotes all trainable parameters in the model. This term penalizes large parameter values, helping to prevent overfitting and improve generalization.
- $\lambda_{\text{temp}} \sum_i \|\mathbf{m}_i^t - \text{stop\_grad}(\mathbf{m}_i^{t-1})\|_2^2$ is the **temporal consistency loss**. It encourages the memory state $\mathbf{m}_i^t$ of each node to evolve smoothly over time by penalizing large changes between consecutive time steps. The stop_grad operation ensures that only the current memory state is updated during backpropagation, treating the previous state as a fixed reference.

Inference: At test time, the model processes each conversation sequentially, updating the dynamic graph structure and temporal memory states for each utterance. Emotion predictions are made for each utterance as the dialogue unfolds, using the most up-to-date context and memory information.

6 Evaluation Metrics

We assess model performance using both quantitative metrics and qualitative analysis:

- **Emotion classification:** Report Macro-F1 and Accuracy to fairly capture performance across all emotions, especially in imbalanced datasets.
- **Emotion shift detection:** Use F1-score and Accuracy to measure how well the model detects changes in emotional state.
- **Ablation studies:** Perform component-wise ablations (e.g., removing speaker indicators, temporal update mechanism or the shift loss term) to quantify each component's contribution.

To assess how well the model captures conversational structure and temporal influence, we conduct a qualitative evaluation of the evolving influence graph. This includes:

- Visualizations that illustrate how edge weights change over time - highlighting which past utterances retain influence or fade.
- Summary statistics, such as distributions of edge weights and how often edges are pruned as the conversation progresses.

Combining these quantitative and qualitative insights enables a comprehensive understanding of both the accuracy of emotion predictions and the effectiveness of dynamic context modeling.

7 Progress So Far

So far, our work has progressed through the following steps:

- We implemented the baseline models, *DialogueRNN* and *DialogueGCN*.
- These baselines were evaluated on the *IEMOCAP* dataset, and their performance was measured on standard metrics (Table 1)
- We trained a *Convolutional Neural Network (CNN)* to handle the utterance encoding stage within our architecture. The loss curve is shown in Figure 1.
- We finalized the subsequent model architecture.

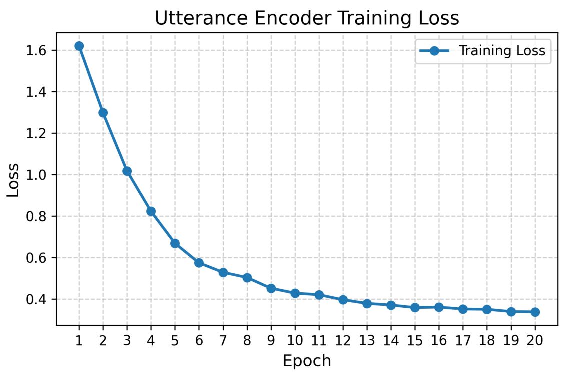

Figure 1: Utterance Encoding CNN Loss Curve

8 Timeline

By the final submission, we expect to have completed the following:

- Development of the dynamic graph update mechanism.
- Experiments and hyperparameter tuning, and creation of visualizations showing graph evolution.
- Final analysis of results and concluding the project.

References

Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N Chang, Sungbok Lee, and Shrikanth S Narayanan. 2008. Iemocap: Interactive emotional dyadic motion capture database. *Language resources and evaluation*, 42(4):335–359.

Deepanway Ghosal, Navonil Majumder, Soujanya Poria, Niyati Chhaya, and Alexander Gelbukh. 2019. Dialoguegcn: A graph convolutional neural network for emotion recognition in conversation. *arXiv preprint arXiv:1908.11540*.

Navonil Majumder, Soujanya Poria, Devamanyu Hazarika, Rada Mihalcea, Alexander Gelbukh, and Erik Cambria. 2019. Dialoguernn: An attentive rnn for emotion detection in conversations. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 6818–6825.

Yuntao Shou, Tao Meng, Wei Ai, and Keqin Li. 2024. Dynamic graph neural ode network for multi-modal emotion recognition in conversation. *arXiv preprint arXiv:2412.02935*.

Yuntao Shou, Jun Yao, Tao Meng, Wei Ai, Cen Chen, and Keqin Li. 2025. Gsdnet: Revisiting incomplete multimodal-diffusion from graph spectrum perspective for conversation emotion recognition. *arXiv preprint arXiv:2506.12325*.

Yan Wang, Bo Wang, Yachao Zhao, Dongming Zhao, Xiaojia Jin, Jijun Zhang, Ruifang He, and Yuexian Hou. 2024. Emotion recognition in conversation via dynamic personality. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 5711–5722.

Table 1: Comparison of DialogueRNN and DialogueGCN on IEMOCAP dataset

Methods	**Happy**		**Sad**		**Neutral**		**Angry**		**Excited**		**Frustrated**		**Average(w)**	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
DialogueRNN	24.23	32.84	74.82	78.80	56.77	58.11	63.15	64.32	80.13	70.97	60.89	57.99	63.07	61.83
DialogueGCN	40.34	42.37	88.73	84.00	61.23	63.05	66.89	63.76	65.01	63.01	63.77	67.01	64.85	63.66